Zoning, Transportation Investment, and Urban Sprawl

Research Proposal Presentation

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Motivation

Zoning and city structure

- Restrictive zoning limits housing supply in desirable area,and forms job-housing separation.
- Households may be pushed toward cheaper but more distant locations.
- This raises commuting costs and sprawing development.

Transportation and city structure

- Rail transit can concentrate development around connected corridors and stations.
- Highway expansion often facilitates decentralization and suburbanization.

Key idea

The effect of transportation infrastructure depends on whether land around accessible locations is allowed to densify.

Research Question

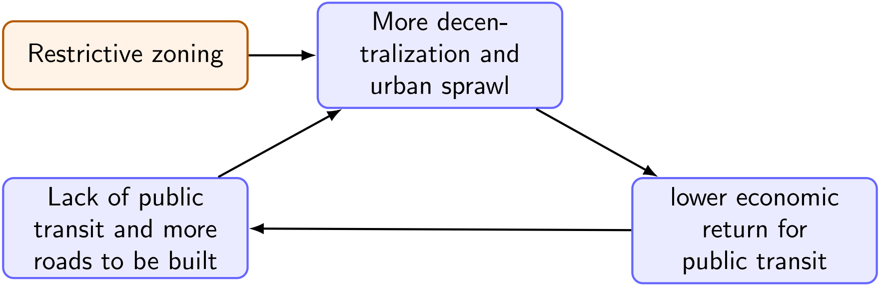
- Restrictive zoning encourage sprawling development
- economic return to public transit is lower in low-density city(**Severen2021-oi**), investment turns towards roadways.
- Lack of public transit cause lower density development
- self-reinforcing...

Central Question

Does the zoning restriction actually have a larger impact if considering its effect on endogeneous public transit investment decisions?

Research Question

For policymakers:



Relation to the Literature

- **Zoning and housing supply: Glaeser2003-sr,Hsieh2019-rxBaum-Snow2024-ii,Rollet2025-mk,Glaeser20**
- **Transportation and congestion externality**:**Severen2021-oi,Allen2025-gj,Baum-Snow2007-zjFajgelbaum2023-dr**
- **Urban sprawl**:**Brueckner2000-vd,Glaeser2003-sr,Coeurdacier2022-ip**

Main Contribution

1. I want to show that the cost and benefit of public transit investment is in fact not exogenous but endogeneous to zoning regulations.
2. How much is the welfare loss by comparing the counterfactual where the zoning regulation is endogenized.

Empirical facts(ongoing)

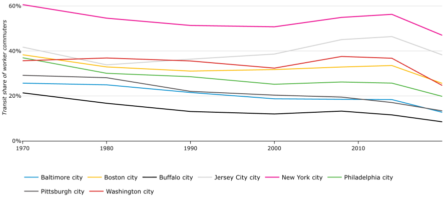
Planned works

1. Transportation investment decisions are driven by the economic returns (cost and benefits)
2. "Old" cities with developed metros before car-dominance have high returns and more transit, wheras "new" cities are oppsite
3. Zoning laws leads to lower return to transit investment.

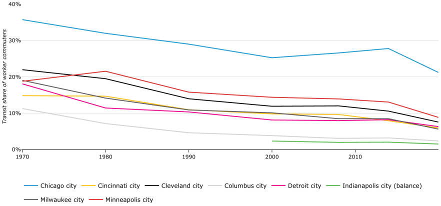
Data source:

- National Transit Database(NTD),Transit Explore Database::"Transit volumnes, revenues and investment, operating date"
- Zoning: housing supply elasticity from **Baum-Snow2024-ii**

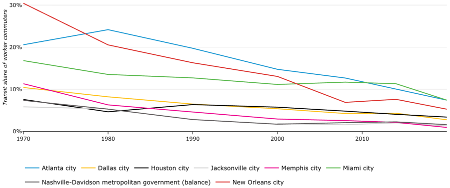
Some Facts



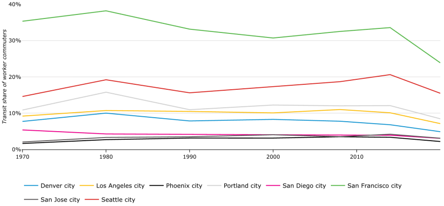
Northeast



Midwest



Southeast



West

Model Environment

Consider a metropolitan area with locations $S = \{1, 2, \ldots, N\}$ partitioned into municipalities $g \in G$.

- Each location has fixed land endowment $\bar{H}_i$.
- Land can be allocated between residential and commercial use:

$$H_i^R + H_i^F \leq \bar{H}_i.$$

- Zoning restricts feasible development intensity in each location.
- Commuting costs depend on transportation infrastructure and congestion(**Bordeu2026-df**):

$$\tau_{ijr} = \tau_{ijr}(I^R, I^T, n).$$

There are three types of agents: households, firms, and governments.

Households and Firms

Households

- Choose residence location i
- Choose workplace location j
- Choose commuting route $r \in R_{ij}$
- Value wages, rents, amenities, and commuting cost

A convenient reduced-form indirect utility is:

$$V_{ijr}(\omega) \propto \frac{B_i}{\tau_{ijr}} \frac{w_j}{(q_i^R)^{1-\beta}} v_{ijr}(\omega).$$

Firms

- Produce a tradable good at workplace location j
- Use labor L_j and commercial land H_j^F

$$Y_j = A_j F(L_j, H_j^F).$$

Competition determines labor demand and commercial land demand.

Household sorting across space depends on wages, rents, amenities, and infrastructure-induced commuting costs.

Government Structure

Municipal governments

- Choose local zoning policies Z_g
- Act as reduced-form representatives of incumbent residents
- Care about local property values, amenities, and congestion/crowding

$$W^g = \sum_{i \in S^g} \left[\omega_1 P_i + \omega_2 B_i^{\text{loc}} - \omega_3 N_i \right],$$

Metropolitan authority

- Chooses road and transit investment
- Internalizes regional transport benefits
- Takes local zoning as given

$$\max \; W^M(I^R, I^T; Z) - C(I^R, I^T),$$

Decentralized Equilibrium and Inefficiency

A decentralized equilibrium features:

- optimal household location, workplace, and route choices,
- optimal firm labor and commercial land choices,
- municipal zoning choices,
- metropolitan road and transit investment,
- market clearing in residential land, commercial land, and labor.

Source of inefficiency

Municipalities restrict development of housing to protect incumbent interests, but do not internalize the broader metropolitan costs in housing supply, commuting, and transit effectiveness.

Efficient Benchmark

Planner's problem

A coordinated planner jointly chooses transportation investment and zoning to maximize aggregate metropolitan welfare.

$$\max_{I^R, I^T, Z} \; W(I^R, I^T, Z) - C(I^R, I^T),$$

Comparing the decentralized equilibrium to this benchmark reveals whether fragmented governance leads to excessive sprawl and underutilized transit-oriented development.

Reference